The empirical recurrence for OEIS A203657, proved by transfer matrix

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3 September 2026

Abstract

OEIS A203657 counts arrays of width 2 over $\{0, 1, \ldots, 7\}$ through two derived arrays of PAIR SUMS — each cell added to its left neighbour, and each cell added to the one above it — on which the entry imposes an order condition. The entry carries an empirical recurrence of order 21 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: the count is a walk count on $S = 128$ states.

2020 Mathematics Subject Classification. 05A15, 06A07, 15A18.

1 The conjecture

OEIS A203657 is "Number of (n+1)X2 0..7 arrays with column and row pair sums b(i,j)=a(i,j)+a(i,j-1) and c(i,j)=a(i,j)+a(i-1,j) such that rows of b(i,j) and columns of c(i,j) are lexicographically nondecreasing.". It has offset 1 and begins

$$1212, 27966, 557544, 9592248, 146861704, 2050339423, 26569429480, 323824032048, \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 70*a(n-1) -2275*a(n-2) +45612*a(n-3) -632289*a(n-4) +6438138*a(n-5) -49938467*a(n-6) +302065016*a(n-7) -1446908771*a(n-8) +5543625850*a(n-9) -17090839633*a(n-10) +42516502924*a(n-11) -85337511267*a(n-12) +137796276966*a(n-13) -177910179993*a(n-14) +181835622416*a(n-15) -144879499352*a(n-16) +87918207776*a(n-17) -39188297232*a(n-18) +12080293632*a(n-19) -2298481920*a(n-20) +203212800*a(n-21)

As of the "Last modified" line on the live entry (Oct 03 2025 10:38:06, revision 10) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array a has 2 cells across, with entries in $\{0, 1, \ldots, 7\}$. Two derived arrays are formed,

$$b(i, j) = a(i, j) + a(i, j - 1) \quad (j \geq 1), \qquad c(i, j) = a(i, j) + a(i - 1, j) \quad (i \geq 1),$$

so b adds each cell to its LEFT neighbour and c adds each cell to the one ABOVE it. The entry asks

that the ROWS of b and the COLUMNS of c are each in lexicographic nondecreasing order. Row i of b is $\big(b(i, 1), \ldots, b(i, 1)\big)$ and column j of c is $\big(c(1, j), c(2, j), \ldots\big)$.

3 Two rows decide it

Lemma 1. *The row comparison is a condition on two consecutive rows of a. The column comparison is settled by the FIRST row in which the two columns of c differ.*

Proof. Row i of b is built from row i of a alone, so comparing rows i and $i+1$ of b reads rows i and $i+1$ of a. Two sequences are compared lexicographically by their first difference; entry i of column j of c is built from rows $i-1$ and i of a, so scanning the array downwards reveals the entries of every column of c in order. □

So the state is the last row of a together with one bit for each adjacent pair of columns, recording whether that pair of columns of c has been equal in every row so far. A row that would put a pair in the wrong order is rejected at once, and a pair still equal at the end is in order, equality being allowed.

With M the adjacency matrix of the digraph on those states, ι the indicator of the starting state and $\tau = \mathbf{1}$,

$$a(n) \;=\; \iota^\top M^j \tau, \qquad S = 128.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 2. *Let $q(t) = t^{21} - \sum_i c_i t^{21-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+21} - \sum_i c_i M^{j+21-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 3.

$$a(n) \;=\; 70\,a(n-1) - 2275\,a(n-2) + 45612\,a(n-3) - 632289\,a(n-4) + 6438138\,a(n-5) - 49938467\,a(n-6) + 3020650$$

for every $n > 20$.

Proof. The vector $w = q(M)\tau$ was formed by 21 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 20 + 1$ onwards and the run continued until the working vector was identically zero. □

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: enumerate every array of that size, form the two derived arrays in full, and test the condition on them directly. No window, no state and no incremental bookkeeping are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A203657.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.